Detector Chains and a Lambert-W Lower Bound for Machine-Dependent Chairman Assignment

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Abstract

For fractional assignments with machine-dependent weights $d_{ij} > 0$, let κ denote the supremum, over all finite instances, of the minimum over integral assignments of the maximum row-and-prefix discrepancy divided by $D = \max_{i,j} d_{ij}$. A companion paper refuted the conjecture $\kappa \le 1$ of Liu and Reis and proved $\kappa \ge 4 - 2\sqrt{2}$ using a three-column detector gadget. Here we replace the three-column detector by a *chain* of $k-1$ detector columns and optimize the resulting family exactly. The optimal chain value admits the closed form

$$\tau(k) \;=\; \max_{d} \frac{d\big(k-1-(k-2)\,d^{1/(k-2)}\big)+1}{1+d}$$

(the maximum over weights keeping the design feasible), and

$$\lim_{k\to\infty} \tau(k) \;=\; 1 + W(1/e) \;\approx\; 1.278465,$$

where W is the Lambert W function; at the optimum the pleasant identity $\tau = 1 + d^*$ holds, with $\ln d^* = -(1+d^*)$. Consequently $\kappa \ge 1 + W(1/e)$. Every rung is certified in exact rational arithmetic: we exhibit explicit instances with certified prefix discrepancy 1.2727 (depth 40), and an independent mixed-integer check confirms a concrete 119×236 instance beating $7/6$. We further report an extensive, exactly verified experimental study of mechanisms beyond chains — banked helpers, bank-manufacturing "shaper" blocks, and block games with forced values above $3/2$ at stated bank states — all of which fail, measurably and instructively, to compose into instances above the chain limit. We conjecture that κ is finite and discuss a scheduler-side program for a matching upper bound, as well as the bounded weight-ratio parameterization $\kappa(\rho)$.

1 Introduction

Let $x \in [0,1]^{m\times n}$ be a fractional assignment, $\sum_i x_{ij} = 1$ for every column j, and let $d \in \mathbb{R}^{m\times n}_{>0}$ be a matrix of machine-dependent weights. For an integral assignment $y \in \{0,1\}^{m\times n}$ with the same column sums, write

$$\Delta_i(t) = \sum_{j=1}^{t} d_{ij}(x_{ij} - y_{ij})$$

for the row-prefix discrepancy. Liu and Reis [2], having proved the common-weight prefix guarantee that resolves a special case of the Morell–Skutella conjecture [4], conjectured that machine-dependent weights always admit y with $|\Delta_i(t)| \le D := \max_{i,j} d_{ij}$ (their Conjecture 19; Conjecture 3 in the arXiv version). The companion paper [5] refuted this with an 11×15 rational instance and strengthened the refutation to a quantitative lower bound: writing

$$\kappa \;=\; \sup_{\text{instances}} \; \min_{y} \; \max_{i,t} \; \frac{|\Delta_i(t)|}{D},$$

it proved $\kappa \geq 4 - 2\sqrt{2} \approx 1.1716$, via a three-column *detector gadget* whose split was optimized, and promised a systematic treatment of deeper detectors. This paper is that treatment.

Results. Our main construction generalizes the three-column detector to a *detector chain* of depth k: one hypothesis column followed by $k-1$ chained detector columns on a fresh row pair, all coupled to a global accumulator row. Optimizing the weights and splits exactly yields three statements.

1. **A closed form per depth** (Theorem 4). The optimal chain of depth k forces prefix discrepancy arbitrarily close to

$$\tau(k) = \max_{d \text{ admissible}} \frac{d\big(k - 1 - (k-2)d^{1/(k-2)}\big) + 1}{1 + d},$$

where admissibility means the balance splits lie in $[0, 1]$ (equivalently the displayed value is at most $1 + d$), and $\tau(3) = 4 - 2\sqrt{2}$ recovers the companion paper's constant.

2. **A Lambert-W limit** (Theorem 5). $\tau(k)$ converges to $1 + W(1/e)$, where $d^* = W(1/e) \approx 0.27846$ satisfies $\ln d^* = -(1 + d^*)$ and the limiting optimal value is exactly $1 + d^*$. Hence $\kappa \geq 1 + W(1/e)$.

3. **Exact certificates** (Section 5). Every claim is machine-verified in exact rational arithmetic. Concrete certified instances include a depth-4 family member ($\tau = 103/85$) whose 119×236 instance provably exceeds $7/6$ (independent mixed-integer infeasibility check) and a depth-40 instance with certified discrepancy ≥ 1.272743, within 0.006 of the limit.

The second half of the paper (Section 6) reports what lies beyond chains. It is experimental in the strict sense that every stated number is an exactly verified fact about a finite object, while the section's larger claims are explicitly conjectural. In brief: block-level mechanisms exist whose forced values exceed $3/2$ *given* favorable bank states, and "shaper" blocks exist that manufacture such states against every legal schedule at fixed budgets; yet every attempt to compose these components into an actual instance has failed, for reasons we can now name and measure (entry pollution, accumulator venting, and the failure of conjunctive bank pinning). The evidence points to a *composition barrier*: the constant κ appears to be governed not by the strength of local gadgets, which is unbounded, but by a corridor-management game that the scheduler keeps winning. We conjecture $\kappa < \infty$ and outline the upper-bound program this suggests.

Why care. Beyond settling the strength of the natural gadget family, the parameterized reading is relevant to scheduling practice: κ prices *unstructured heterogeneity* in prefix-fair scheduling. Section 7 records the bounded-ratio corollary: with all weights in $[D/\rho, D]$, the constructions degrade gracefully, $\kappa(\rho) \to 1 + W(1/e)$ as $\rho \to \infty$, while $\kappa(1) = 1$ by the common-weight theorem; the curve between is open and is, in our view, the right target for applications.

2 The chain gadget

Fix a depth $k \geq 3$, a detector share $p \in (0, 1)$, an off-support weight $\eta > 0$, and a block count N. The rows are $B, A_1, H_1, \ldots, A_N, H_N$; the columns come in N consecutive blocks

$$J_b,\ C_2^b,\ C_3^b,\ \ldots,\ C_k^b \qquad (b = 1, \ldots, N).$$

Within block b (write $A = A_b$, $H = H_b$), the nonzero fractional entries and exceptional weights are:

column	x_A	$x_\cdot$	d_A	$d_\cdot$	
J	$1-p$	$p\ (B)$	1	1	
C_j	x_j	$1-x_j\ (H)$	d_j	1	$(j = 2, \ldots, k)$

with weights $d_2 \le \cdots \le d_{k-1} \le d_k = 1$ and splits $x_j \in [0,1]$ to be chosen; every unlisted weight equals η. All weights are strictly positive and $D = 1$.

The forcing logic generalizes the three-column case. Suppose a hypothetical assignment keeps all prefixes within a budget c, and suppose it denies J to A. Then A sits near $1-p$, and each chain column presents the scheduler with a dilemma: assigning C_j to A relieves A by $d_j(1-x_j)$ but denies H its share $1-x_j$, while any other choice denies A the amount $d_j x_j$ and fires immediately. The weights grow along the chain, so the relief purchased early is small while the punishment for late deviation stays high; the terminal column finally forces a violation on one of the two rows.

Lemma 1 (block heights)**.** *Deny J to A and let the scheduler play the block columns arbitrarily. With the convention that sums over i range over chain columns before the current one, every continuation attains one of the following prefix heights on A or H:*

$$\begin{aligned} \mathrm{dev}_j &= (1-p) - \textstyle\sum_{i<j} d_i(1-x_i) + d_j x_j && (\textit{first deviation at } C_j,\ j<k), \\ \mathrm{term}_A &= (1-p) - \textstyle\sum_{i<k} d_i(1-x_i) + x_k && (\textit{terminal denied to } A), \\ \mathrm{term}_H &= \textstyle\sum_{i<k}(1-x_i) + (1-x_k) && (\textit{full compliance}). \end{aligned}$$

These heights are exact up to the η-bookkeeping of Lemma 2: a column sent to neither support row denies both, so it only raises the listed heights.

Proof. If the first chain column not assigned to A is C_j with $j<k$, then at that prefix A has received exactly the earlier chain columns and been denied J's share and C_j's share, giving dev_j; this is attained at the prefix of C_j itself, before any later column can repair it. If all of $C_2, \ldots, C_{k-1}$ go to A and C_k does not, A's height at the terminal prefix is term_A. If every chain column goes to A, then H has been denied its share $1-x_j$ of every chain column, and its height at the terminal prefix is term_H. Assignments off the support deny both rows and can only increase these values (they subtract nothing and cost the recipient only η, which Lemma 2 accounts). □

Lemma 2 (generalized budget lemma)**.** *Let τ be a value such that the $p=0$ idealizations of the heights of Lemma 1 (the same expressions with leading term 1 in place of $1-p$) are all at least τ. Then every integral assignment of the N-block instance has some row-prefix discrepancy at least*

$$\mathrm{LB} \ = \ \min\Big\{\ \tau - p - \big(k(N-1)+1\big)\eta, \quad Np - N(k-1)\eta\ \Big\}.$$

Proof. The fresh rows of block b appear off support in at most $k(N-1)$ earlier columns, each costing η, so they enter the block no lower than $-k(N-1)\eta$. Within the block, A is on support in every column, so it takes no further η hits; H can additionally absorb J off support once. The heights of Lemma 1 lose exactly p on the A-side entries (their first term is $1-p$ rather than 1) and nothing on term_H; combining, every height is at least $\tau - p - (k(N-1)+1)\eta$. If this exceeds the running budget, denying any J_b is impossible, so all N detectors are assigned to their A_b; then B is denied its share p in every block and can be relieved only by off-support assignments of chain columns, each worth η, of which there are at most $N(k-1)$. At the final prefix $\Delta_B \ge Np - N(k-1)\eta$. Applying the argument with the budget equal to the assignment's actual maximum prefix discrepancy yields the claim. □

3 Optimizing the chain

Equalize the heights. In the idealization $p \to 0$, setting all dev_j, term_A, term_H equal to a common value τ forces the *balance solution*

$$x_{j+1} = \frac{d_j}{d_{j+1}} \quad (2 \le j \le k-1,\ d_k := 1), \qquad x_2 = \frac{\tau - 1}{d_2}, \tag{1}$$

as one checks by subtracting consecutive height expressions. Substituting into $\mathrm{term}_H = \tau$ and writing

$$R \;=\; \sum_{j=2}^{k-1} \frac{d_j}{d_{j+1}}$$

gives $(k-1) - \frac{\tau-1}{d_2} - R = \tau$, that is,

$$\tau \;=\; \frac{d_2\,(k-1-R)+1}{1+d_2}. \tag{2}$$

Lemma 3 (geometric weights are optimal)**.** *For fixed d_2, the value* (2) *is maximized when $d_2, \dots, d_{k-1}, 1$ is a geometric progression, in which case $R = (k-2)\,d_2^{1/(k-2)}$.*

Proof. The ratios $r_j = d_j/d_{j+1}$ satisfy $\prod_{j=2}^{k-1} r_j = d_2/d_k = d_2$. By AM–GM, $R = \sum r_j \ge (k-2)\big(\prod r_j\big)^{1/(k-2)} = (k-2)d_2^{1/(k-2)}$, with equality iff all ratios are equal. Since (2) is decreasing in R, the geometric choice is optimal. □

Call a weight $d \in (0,1]$ *admissible* for depth k if the geometric balance design has all splits in $[0,1]$. The interior ratios equal $d^{1/(k-2)} \le 1$ automatically, so admissibility is exactly the condition $x_2 = (\tau-1)/d \le 1$, that is, $\tau(k,d) \le 1+d$, where

$$\tau(k,d) \;:=\; \frac{d\big(k-1-(k-2)\,d^{1/(k-2)}\big)+1}{1+d}.$$

(Small d can be inadmissible at large k; every table point of Section 5 is verified admissible.)

Theorem 4 (closed form)**.** *For every $k \ge 3$,*

$$\tau(k) \;=\; \max_{d \text{ admissible}} \frac{d\big(k-1-(k-2)\,d^{1/(k-2)}\big)+1}{1+d},$$

and for every rational admissible point with strict margins and every $\varepsilon > 0$, suitable (p, η, N) in Lemma 2 give instances forcing discrepancy $\ge \tau(k,d) - \varepsilon$ with $D = 1$. In particular $\kappa \ge \tau(k,d)$ for every admissible pair.

Proof. Combine (2) with Lemma 3 and write $d = d_2$; admissibility is precisely feasibility of the design. For the quantitative statement take $p = \varepsilon/3$, $N = \lceil(\tau+1)/p\rceil$, and $\eta = \varepsilon/(3kN)$: then $\tau - p - (k(N-1)+1)\eta > \tau - \varepsilon$ and $Np - N(k-1)\eta > \tau$, so Lemma 2 applies. □

4 The limit is a Lambert-W constant

Theorem 5.

$$\lim_{k\to\infty} \tau(k) \;=\; 1 + W(1/e),$$

where $d^* = W(1/e)$ is the unique root of $\ln d = -(1+d)$ in $(0,1)$. Consequently $\kappa \ge 1 + W(1/e) \approx 1.2784645$.

Proof. Write $m = k-2$ and $\phi_m(d) = 1+m\big(1-d^{1/m}\big)$, so that $\tau(k,d) = [d\,\phi_m(d)+1]/(1+d)$. For fixed $d \in (0,1)$ the quantity $m(1-d^{1/m}) = m(1-e^{\ln d/m})$ increases in m to $-\ln d$ (apply $e^x \geq 1+x$ with $x = \ln d/m$), so

$$\tau(k,d)\ \uparrow\ \tau_\infty(d) := \frac{d(1-\ln d)+1}{1+d} \qquad (k\to\infty).$$

Differentiating τ_∞, the numerator of the derivative is

$$(1+d)\frac{\partial}{\partial d}\big[d - d\ln d + 1\big] - \big[d(1-\ln d)+1\big] \ = \ -\ln d - d - 1,$$

which decreases from $+\infty$ (as $d \downarrow 0$) to -2 (at $d=1$); so τ_∞ increases on $(0,d^*]$ and decreases on $[d^*,1]$, where $\ln d^* = -(1+d^*)$, equivalently $d^* e^{d^*} = 1/e$, i.e. $d^* = W(1/e)$ [1]. Substituting $\ln d^* = -(1+d^*)$,

$$\tau_\infty(d^*) = \frac{d^*(1+1+d^*)+1}{1+d^*} = \frac{(1+d^*)^2}{1+d^*} = 1+d^*.$$

Upper bound. For any k and any admissible d we have $\tau(k,d) \leq \tau_\infty(d) \leq \tau_\infty(d^*) = 1+d^*$, hence $\tau(k) \leq 1+d^*$ for every k.

Lower bound. Fix $d \in (d^*,1)$. Then $\tau_\infty(d) \leq 1+d^* < 1+d$, so $\tau(k,d) < 1+d$ for *every* k: the pair (k,d) is always admissible. Since $\tau(k,d) \uparrow \tau_\infty(d)$, we get $\liminf_k \tau(k) \geq \tau_\infty(d)$, and letting $d \downarrow d^*$ (where τ_∞ is continuous) gives $\liminf_k \tau(k) \geq 1+d^*$. The two bounds give the limit. Finally $\kappa \geq \tau(k,d)$ for every admissible pair by Theorem 4, so $\kappa \geq \tau_\infty(d)$ for every $d > d^*$, and $\kappa \geq 1+d^*$ follows. □

Remark 6. The identity $\tau_\infty = 1+d^*$ says: at the optimum, the limiting chain forces exactly one plus its own starting weight. We do not know a direct combinatorial explanation; it drops out of the stationarity condition.

5 Exact certificates

All claims above are verified by machine in exact rational arithmetic. Two independent toolchains are used: a certificate checker that re-derives every inequality of Lemmas 1 and 2 for a given (k,d,p,η,N) from scratch using exact fractions, and, for one concrete family member, an independent mixed-integer feasibility check of the assembled instance.

depth	weights	τ (exact)	τ (dec.)	certified instance LB
$k=3$	$d=5/12$	239/204	1.171569	1.161469
$k=4$	$d=(9/25,3/5)$	103/85	1.211765	1.201665
$k=5$	$d=(1/3,10/21,7/10)$	904/735	1.229932	1.219832
$k=7$	near-geometric	211933/170170	1.245419	1.235320
$k=12$	$r=22/25$	(exact rational)	1.261401	1.251301
$k=25$	$r=473/500$	(exact rational)	1.270876	1.269776
$k=40$	$r=967/1000$	(exact rational)	1.273843	**1.272743**

Here r denotes the common ratio of a rational geometric chain $d_j = r^{k-j}$, which keeps every balance split (1) rational ($x_{j+1} = r$). The certified instance lower bounds instantiate Lemma 2 with $p = 1/100$, $N = 300$ for $k \leq 12$ and $p = 1/1000$, $N = 3000$ for the deep rungs, $\eta = 1/(10^4 kN)$ throughout; each is an explicit finite instance with $D = 1$ and every column of fractional support two. The depth-4 member with $p = 1/50$, $N = 59$ ($m = 119$, $n = 236$) was additionally checked by an exact mixed-integer solver: infeasible at budget 7/6, feasible at 1.19. This certifies the promise made in the companion paper that deeper detectors beat $4-2\sqrt{2}$: already $k=4$ reaches $103/85 > 4-2\sqrt{2} > 7/6$.

6 Beyond chains: what is true, and what fails

Everything in this section is an exactly verified statement about finite *block games*, together with an honest account of why none of it currently yields instances above the chain limit. A block game fixes a bank state (starting heights on designated rows), a budget c, and a block of columns; all quantifiers over schedules are discharged by exhaustive search in exact arithmetic.

Banked chains. If the helper row enters a chain block at height $h \geq 0$, the analysis of Sections 2–3 goes through with term_H increased by h, giving

$$\tau(k,h) = \max_{d \text{ adm.}} \frac{d\big(k-1+h-(k-2)d^{1/(k-2)}\big)+1}{1+d}, \qquad \lim_{k} \tau(k,h) = 1 + W(e^{h-1}),$$

and on the compliant path the helper ends at exactly $\tau - 1$. Iterating $h \mapsto \tau(\cdot, h) - 1$ has fixed point $h^* = 1/e$, suggesting a bound of $1 + 1/e \approx 1.3679$. *This is not a theorem*, and our experiments show why the natural proof fails: sustaining h across blocks requires a guarantee over *all* legal schedules, and a single legal gift of a terminal column to the helper resets its bank from h to $h-1$. The exists-a-schedule bookkeeping that suffices for one block does not compose.

Super-threat blocks and shapers (exact). Two block-game facts, each verified by exhaustive exact search over all schedules (4^n plays, pruned): (i) there is an explicit 14-column block that, entered at bank state $(1/2, 9/20)$ on its two designated rows and with its detector column denied, forces every schedule to a prefix height of exactly $1.687669\ldots > 3/2$, while admitting a legal escape play of value 0.6931 when the detector is granted; (ii) there are *cold shaper* blocks with no entry banks at all such that, at every budget $c \leq 1.35$, every c-legal schedule exits with some designated row at height ≥ 0.3088. Further stage games, each verified separately at its stated entry state, lift such banks in exact steps (for instance, at budget 1.48, from 0.31 through 0.55), and a bank of 0.6588 fires exactly: the denial of a unit detector from that height is worth 1.63 on every play, exceeding the budget 1.60 at which the corresponding stage guarantees were verified. We emphasize that these are separate block games, each at its stated entry state; whether they can be made to occur in sequence inside one instance is precisely the composition question below. At the level of components, however, no budget looks safe: for every c we found stage games whose guarantees, *if composable*, would force a violation beyond c.

The composition barrier (measured). None of this composes. In a fixed instance the scheduler plays all blocks on shared rows, and three exactly measured mechanisms defeat every assembly we tried. *Entry pollution*: stage guarantees proved at a bank state are vacuous when the scheduler arrives elsewhere in the $\pm c$ corridor, which it can, having parked rows anywhere within budget. *Accumulator venting*: when the accumulator nears the budget, one sacrificial receipt (a detector column granted to the accumulator itself) vents it by nearly a full weight, and pre-parking the pressured row makes the accompanying denial spike harmless; in our assembled 1273-column machine the accumulator was observed climbing to 1.279 and venting to 0.359. *No conjunctive pinning*: exhaustive search over our block class shows no block can force *both* designated rows to exit above useful thresholds against all legal plays (best exact deficits -0.32 to -0.51); exits are disjunctive, so the scheduler always has a coordinate to wreck. The bottom line is quantitative: the assembled machine built from components with stage-level guarantees at $c = 1.35$ admits a schedule with maximum prefix discrepancy ≤ 1.116 — weaker than the plain depth-4 chain instance. Block strength is unbounded; composition is the constraint.

Conjecture and program.

Conjecture 7. $\kappa < \infty$.

Two further data points support this. First, a simple online rule (relieve the taller support row unless doing so would sink it below a fixed floor; otherwise relieve the other; otherwise dump on the row with most headroom) already achieves maximum prefix discrepancy 1.116 on our strongest assembled instance and within 0.004 of the exact optimum on the companion paper's instances; its worst case over adversarial search is below 2.2 on everything we could construct. Second, the mechanisms that defeat composition — corridor slack, venting, disjunctive exits — are exactly the ingredients a potential-function upper-bound proof would formalize: the scheduler maintains every row in a managed corridor and can always afford one sacrificial relief before any accumulator bursts. Adapting the earliest-deadline analysis of [2] with row-specific deadlines and such a potential is, in our view, the most promising route to any finite upper bound; even $\kappa \leq O(1)$ would be the first such statement for machine-dependent weights. In the other direction, any instance forcing more than $1 + W(1/e)$ must defeat the corridor mechanisms simultaneously on all rows, which our experiments consistently failed to arrange.

7 Bounded weight ratio

For applications the relevant parameter is often the dynamic range of the weights rather than their absolute scale. Define $\kappa(\rho)$ as κ restricted to instances with $d_{ij} \in [D/\rho, D]$. Here $\kappa(1) = 1$: the upper bound is the common-weight theorem of Liu and Reis, and Tijdeman's classical lower bound $1 - 1/(2m-2) - \delta$ supplies the matching supremum [3, 6]. Our constructions use tiny off-support weights, but only through the bookkeeping of Lemma 2: setting $\eta = 1/\rho$ there gives, for any admissible depth-k point with value τ and any p,

$$\kappa(\rho) \;\geq\; \min\Big\{\tau - p - \tfrac{k(N-1)+1}{\rho},\; Np - \tfrac{N(k-1)}{\rho}\Big\},$$

and choosing $N = \lceil 2/p \rceil$, $p = 2\sqrt{k/\rho}$ yields $\kappa(\rho) \geq \tau(k,d) - 4\sqrt{k/\rho}$ once $\rho \geq 16k$. Since $\sup_{k,d} \tau(k,d) = 1 + W(1/e)$, it follows that $\kappa(\rho) \to 1 + W(1/e)$ as $\rho \to \infty$; for a concrete data point, $\rho \geq 3.2 \times 10^4$ already gives $\kappa(\rho) > 7/6$ via the depth-4 witness. We make no attempt to optimize the rate, which is certainly far from tight. Determining the shape of $\kappa(\rho)$ between $\kappa(1) = 1$ and the machine-dependent regime — mild heterogeneity versus mixed fleets, in scheduling terms — is open and seems the right quantitative target for practice.

8 Open problems

1. Is κ finite (Conjecture 7)? Any finite upper bound would be the first for machine-dependent prefix discrepancy.

2. Is $\kappa = 1 + W(1/e)$? The chain limit is the strongest instance-level lower bound we know, all mechanisms beyond it having failed to compose; but we see no conservation obstruction above it, and the block-game results show local strength is not the bottleneck.

3. The chain family is optimal among single-pair detector chains by construction; is it optimal among all single-block gadgets? Our search evidence (some 16,000 mixed-integer evaluations over tree-shaped punishment structures at depths 4–7, spot-validated against exhaustive exact search) found nothing better, but this is not a proof.

4. Determine $\kappa(\rho)$ asymptotics and, for practice, its profile at moderate ρ.

Reproducibility and disclosure

The accompanying repository contains exact verifiers for every inequality in Lemmas 1 and 2 at all seven table rungs, including the deep geometric chains (`Fraction`-exact throughout); the mixed-integer confirmation of the 119×236 instance; the exhaustive block-game verifiers behind Section 6 (branch-and-bound over all schedules in scaled integer arithmetic); and the experiment logs, including documented retractions of intermediate findings that deeper search or exact re-verification overturned. Generative AI systems (GPT-5.6 Pro and Claude Fable 5) assisted with exploration, computation, verification code, and exposition. They are not authors. The author assumes responsibility for independent verification, priority, attribution, and the final text.

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